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Section:BSCS-4K

Lab Task:9

Task:1

Code:

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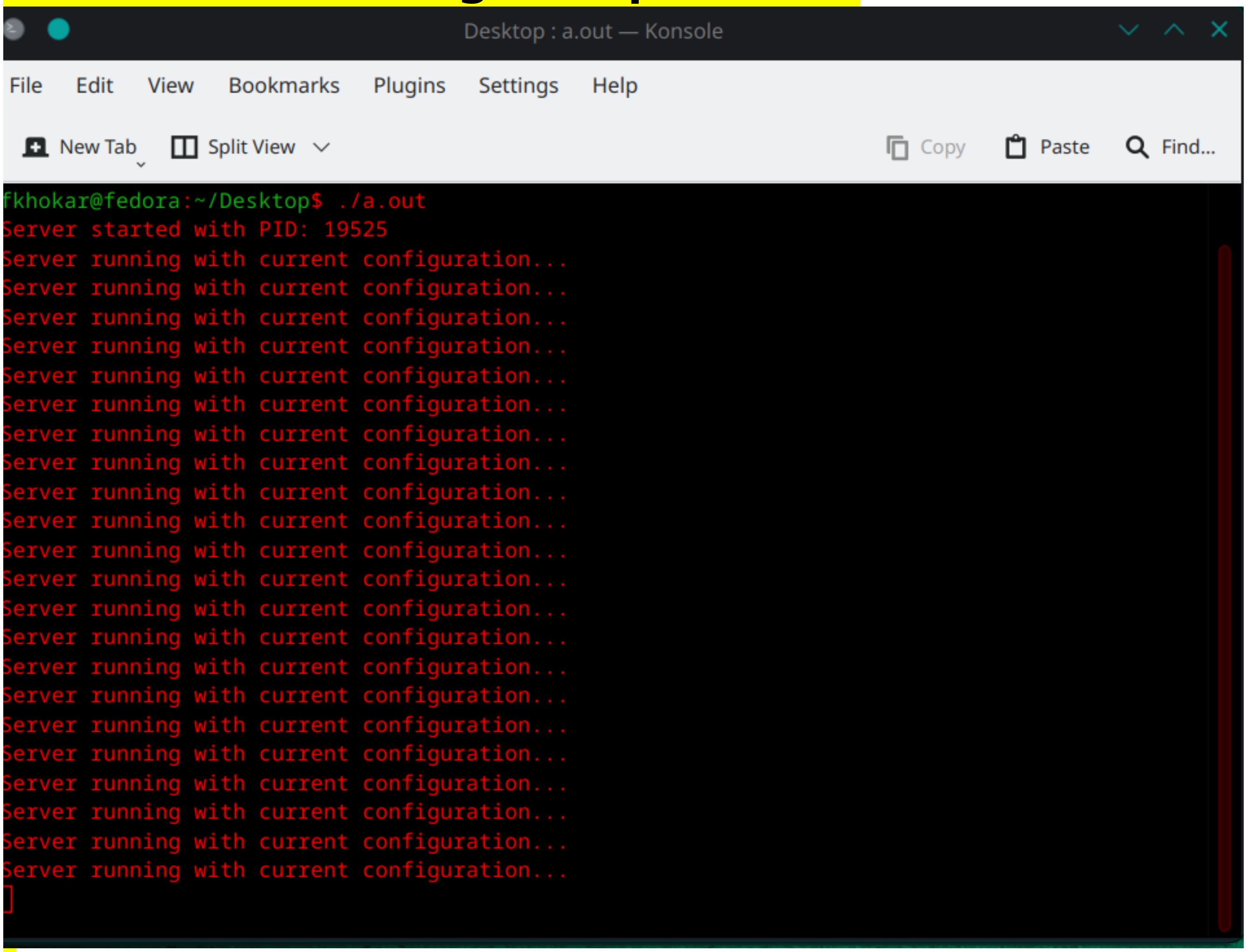
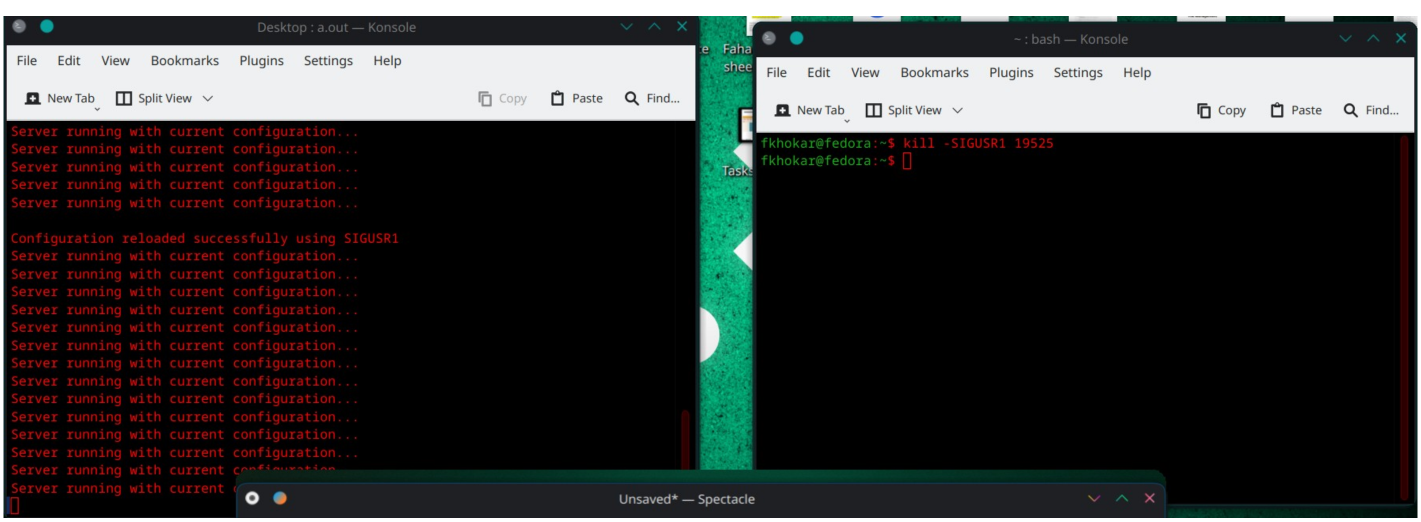
```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>


void handle_sigusr1(int sig) {
    if (sig == SIGUSR1) {
        printf("\nConfiguration reloaded successfully using SIGUSR1\n");
        fflush(stdout);
    }
}

int main() {

    if (signal(SIGUSR1, handle_sigusr1) == SIG_ERR)     // this is handling Register signal handler......
    {
        perror("Error setting signal handler");
        exit(EXIT_FAILURE);
    }

    printf("Server started with PID: %d\n", getpid());

    // this shoow simulation server work continouslyyy.........
    while (1) {
        printf("Server running with current configuration...\n");
        fflush(stdout);
        sleep(2);
    }

    return 0;
}
```

Output:

now serevr running with pid 19525



After sending signal from other termina...



Task:2
code:

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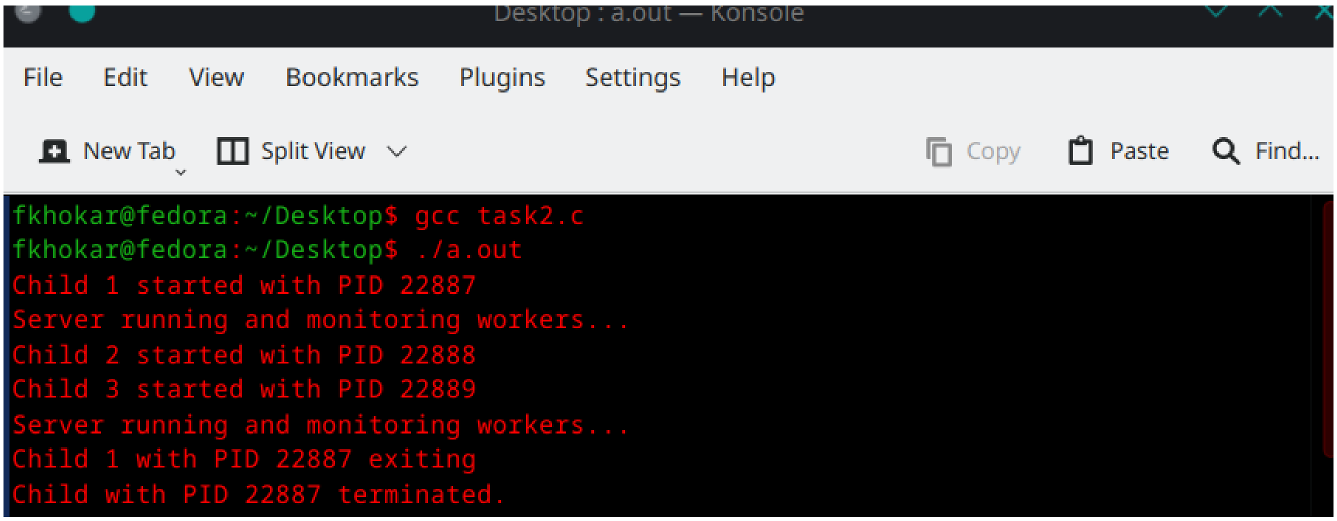
task1.c task2.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

#define NUM_CHILDREN 3
void sigchld_handler(int sig) {
    int status;
    pid_t pid;

    while ((pid = waitpid(-1, &status, WNOHANG)) > 0) { // this termiante terminate the children
        printf("Child with PID %d terminated.\n", pid);
    }
    fflush(stdout);
}
int main() {
    pid_t children[NUM_CHILDREN];
    int sleep_times[NUM_CHILDREN] = {5, 7, 10};   //this is wait time if chile...


    signal(SIGCHLD, sigchld_handler);


    for (int i = 0; i < NUM_CHILDREN; i++)            // this creating children
    {
        pid_t pid = fork();
        if (pid == 0) {
            printf("Child %d started with PID %d\n", i+1, getpid());
            sleep(sleep_times[i]);
            printf("Child %d with PID %d exiting\n", i+1, getpid());
            exit(0);
            } else {
            children[i] = pid;
                }
    }
```

```c
            children[i] = pid;
                }
    }
    while (1){
        printf("Server running and monitoring workers...\n");
        fflush(stdout);
        sleep(3);
    }

    return 0;
}
```

output:

afterr running



After killing from other terminal